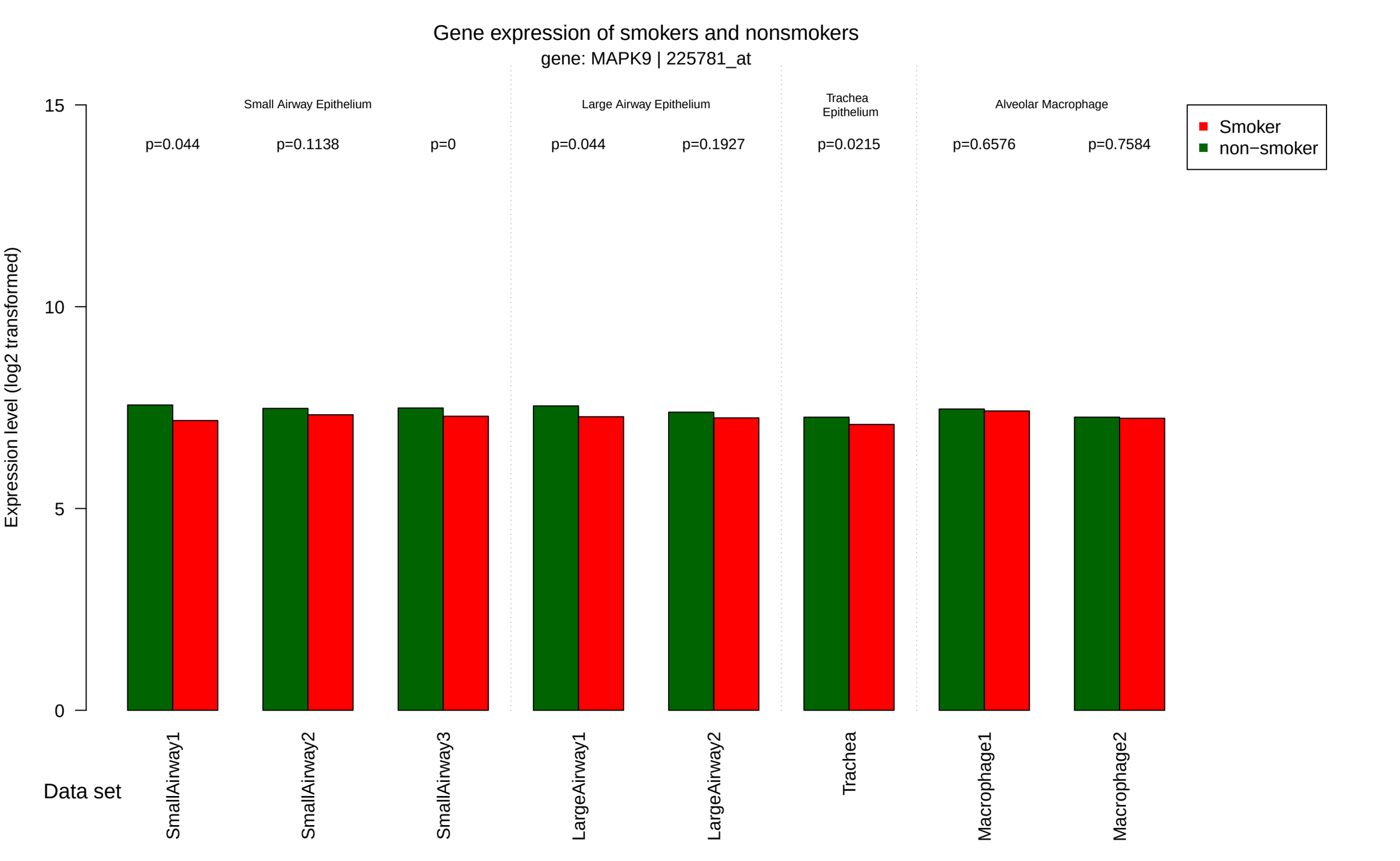
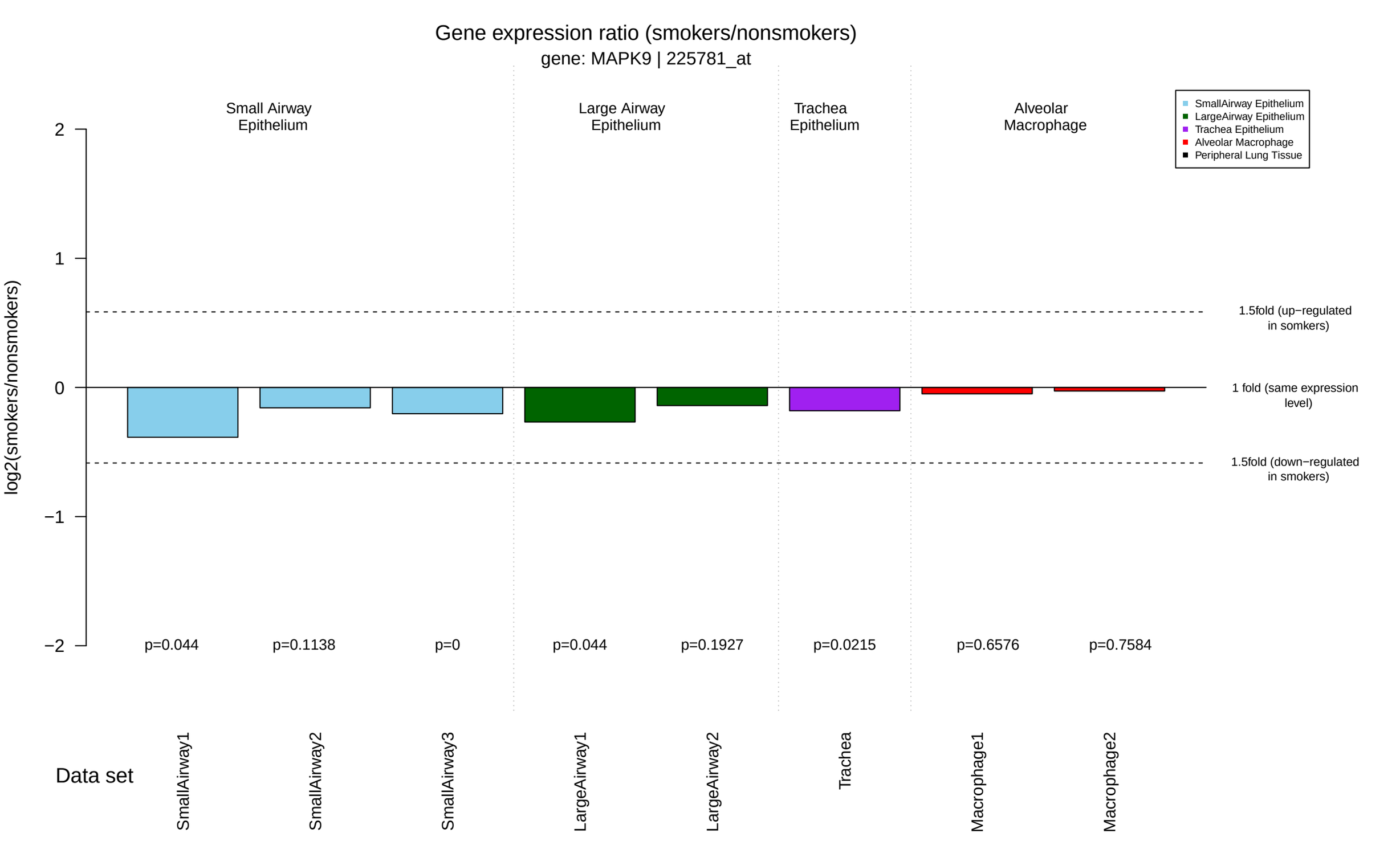
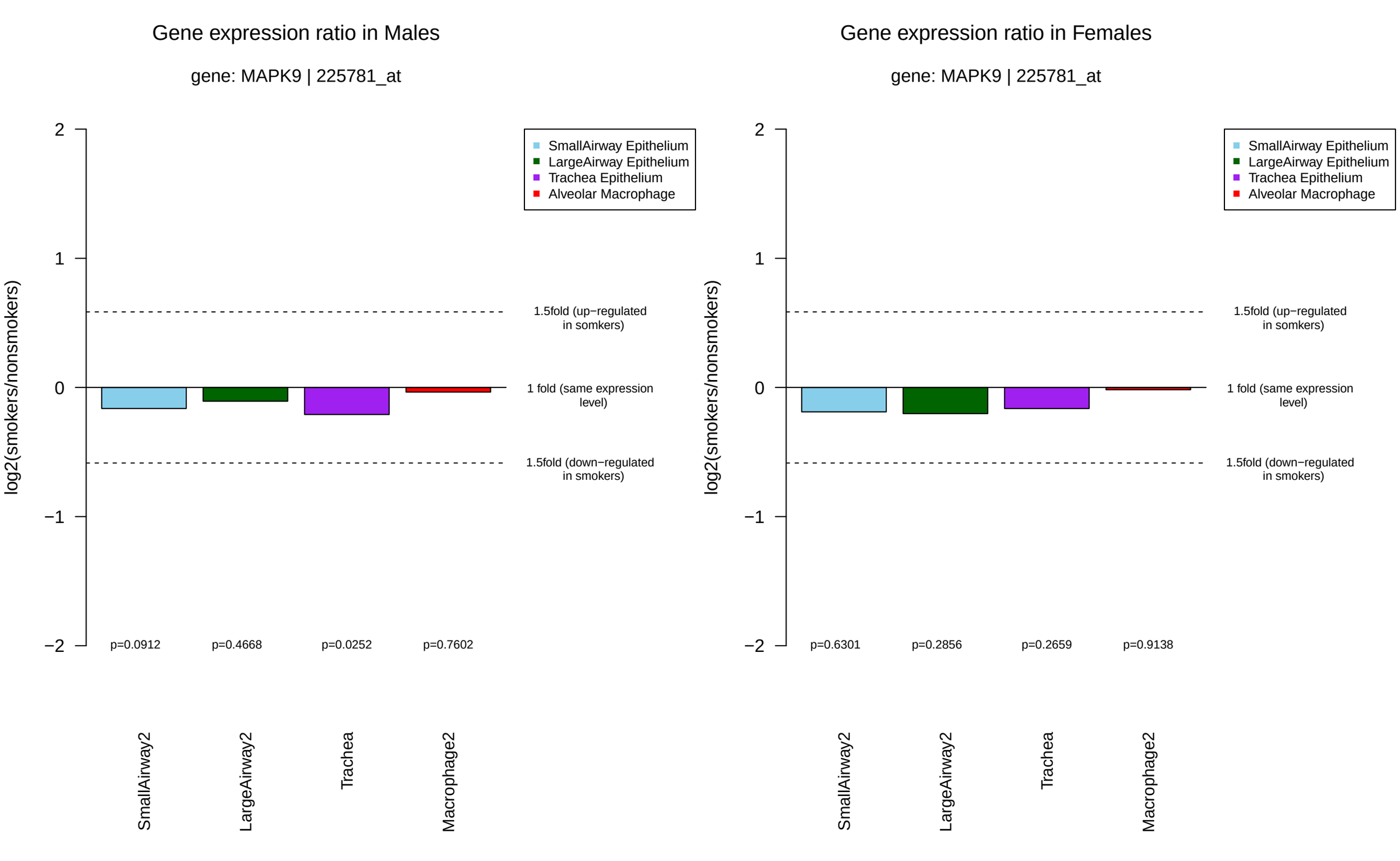
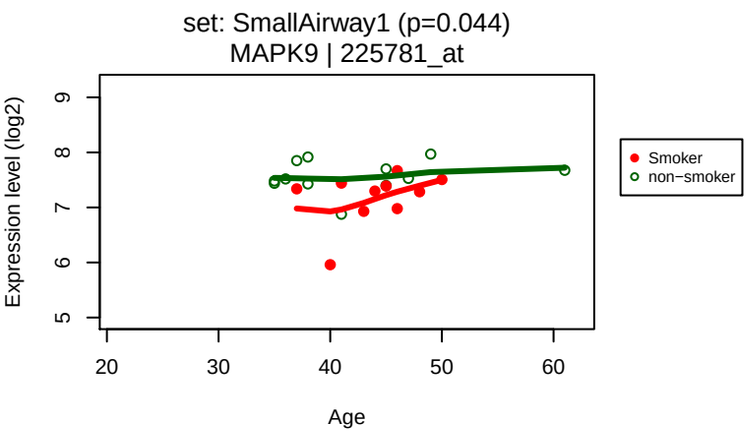
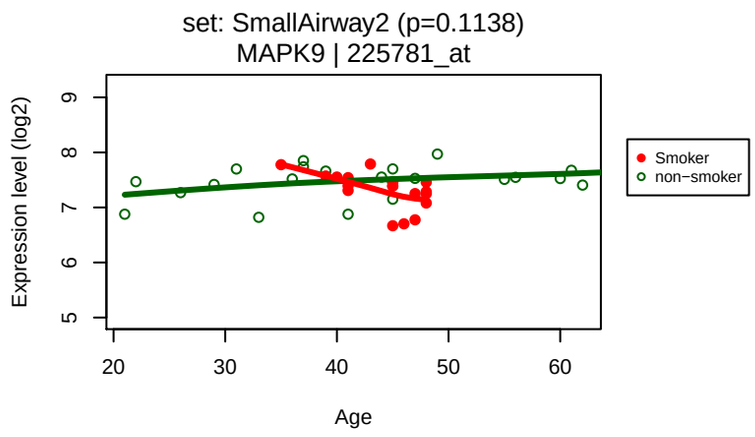
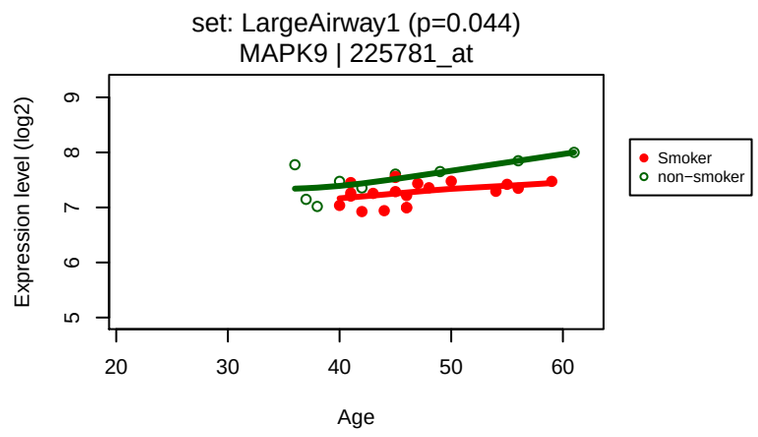

Gene expression of smokers and nonsmokers
gene: MAPK9 | 225781_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.044
p=0.1138
p=0
p=0.044
p=0.1927
p=0.0215
p=0.6576
p=0.7584
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

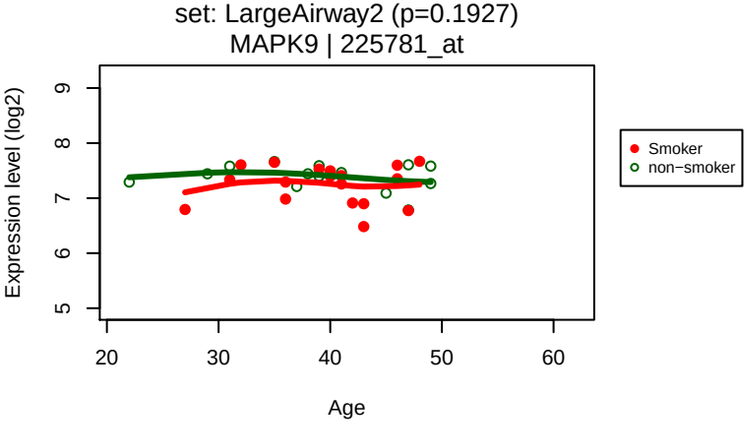

Gene expression ratio (smokers/nonsmokers)
gene: MAPK9 | 225781_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.044
p=0.1138
p=0
p=0.044
p=0.1927
p=0.0215
p=0.6576
p=0.7584
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

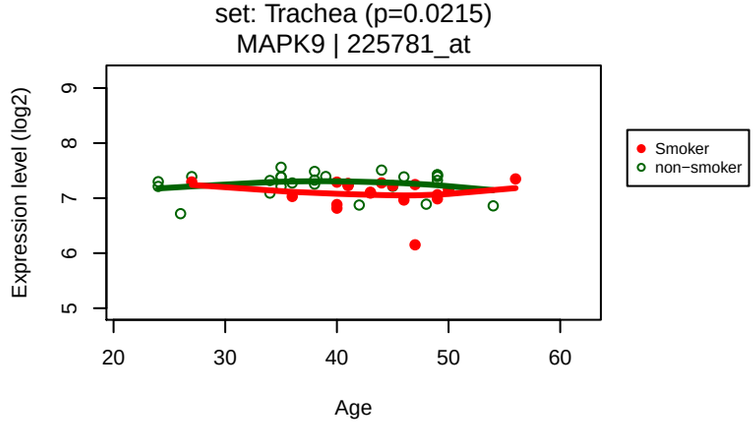

Gene expression ratio in Males
gene: MAPK9 | 225781_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0912
p=0.4668
p=0.0252
p=0.7602
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK9 | 225781_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6301
p=0.2856
p=0.2659
p=0.9138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

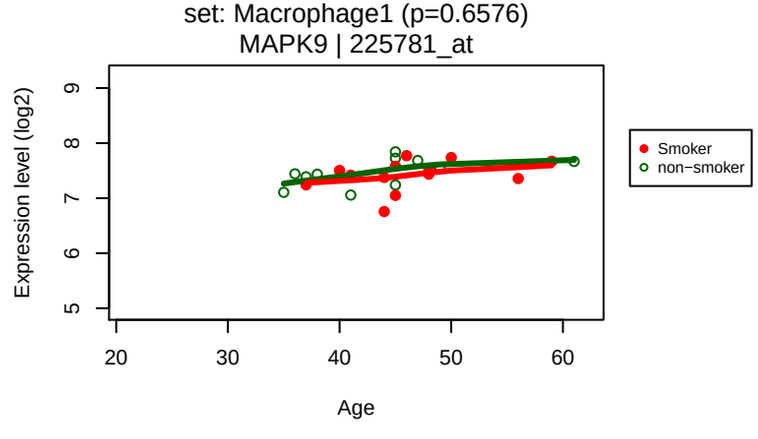

set: SmallAirway1 (p=0.044)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker

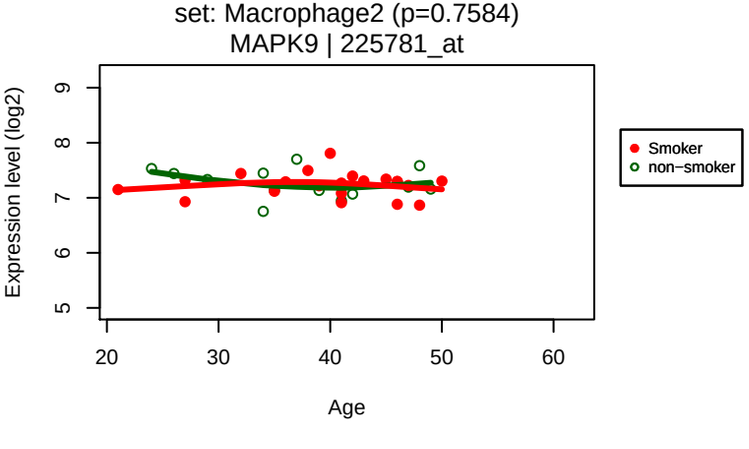

set: SmallAirway2 (p=0.1138)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker

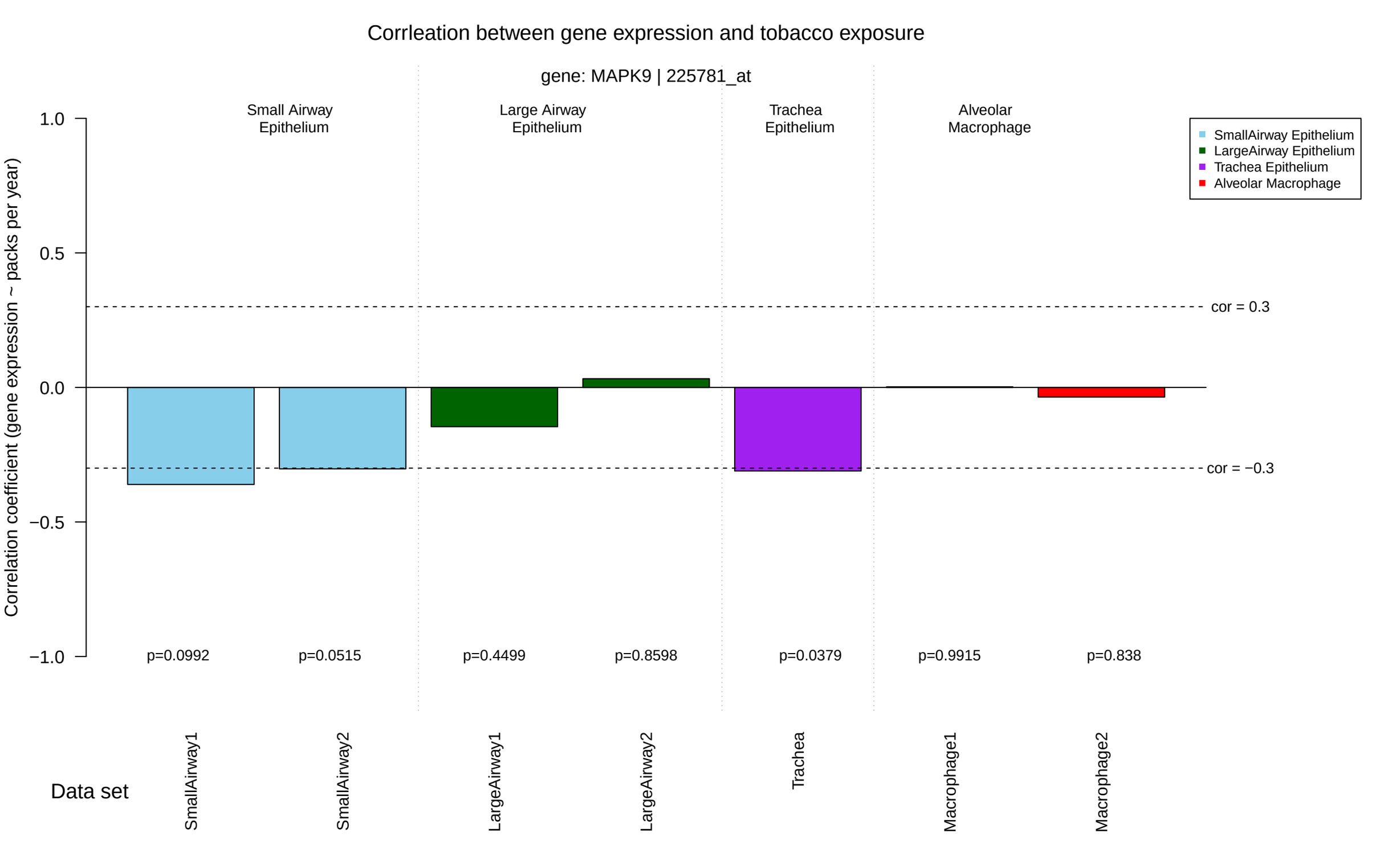

set: LargeAirway1 (p=0.044)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1927)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0215)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6576)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7584)
MAPK9 | 225781_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK9 | 225781_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0992
p=0.0515
p=0.4499
p=0.8598
p=0.0379
p=0.9915
p=0.838
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2